



Mark Scheme (Results)

Summer 2024

Pearson Edexcel GCE
In Mathematics (9MA0)
Paper 31 Statistics

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Summer 2024

Question Paper Log Number 74093

Publication Code 9MA0_01_2406_MS

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General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

EDEXCEL GCE MATHEMATICS

General Instructions for Marking

1. The total number of marks for the paper is 50.
2. The Edexcel Mathematics mark schemes use the following types of marks:
 - **M** marks: method marks are awarded for 'knowing a method and attempting to apply it', unless otherwise indicated.
 - **A** marks: Accuracy marks can only be awarded if the relevant method (M) marks have been earned.
 - **B** marks are unconditional accuracy marks (independent of M marks)
 - Marks should not be subdivided.

3. Abbreviations

These are some of the traditional marking abbreviations that will appear in the mark schemes.

- bod – benefit of doubt
- ft – follow through
- the symbol $\surd$ will be used for correct ft
- cao – correct answer only
- cso - correct solution only. There must be no errors in this part of the question to obtain this mark
- isw – ignore subsequent working
- awrt – answers which round to
- SC: special case
- oe – or equivalent (and appropriate)
- dep – dependent
- indep – independent
- dp decimal places
- sf significant figures
- * The answer is printed on the paper
- ☐ The second mark is dependent on gaining the first mark

4. For misreading which does not alter the character of a question or materially simplify it, deduct two from any A or B marks gained, in that part of the question affected.
5. Where a candidate has made multiple responses and indicates which response they wish to submit, examiners should mark this response.
If there are several attempts at a question which have not been crossed out, examiners should mark the final answer which is the answer that is the most complete.

6. Ignore wrong working or incorrect statements following a correct answer.
7. Mark schemes will firstly show the solution judged to be the most common response expected from candidates. Where appropriate, alternative answers are provided in the notes. If examiners are not sure if an answer is acceptable, they will check the mark scheme to see if an alternative answer is given for the method used.

Qu 1	Scheme	Marks	AO
(a)	$X \sim B(10, \frac{1}{6})$ [Allow 0.167 or better for $\frac{1}{6}$]	M1	3.3
(i)	$[P(X = 3) =]$ 0.155045... awrt 0.155	A1	1.1b
(ii)	$[P(X < 3) = P(X \leq 2) =]$ 0.775226... awrt 0.775	A1	1.1b
		(3)	
(b)	[Let D = no. of days when $X = 3$] $D \sim B(60, "0.155")$	M1	3.3
	$P(D \geq 12) = 1 - P(D \leq 11)$ [Allow $1 - P(D < 12)$]	M1	3.4
	$= 1 - 0.78819...$ awrt 0.212	A1	1.1b
		(3)	
(c)	$[n = 600, p = \frac{1}{6}]$ estimate = 100	B1	3.4
		(1)	
(d)	$[S = \text{total no. of sixes over 60 days.}] S \approx T \sim N\left("100", \sqrt{\frac{5}{6} \times 100}^2\right)$	M1A1	3.3, 1.1b
	$P(S > 95) \approx P([T >] 95.5)$ or $P\left([Z >] \frac{95.5 - "100"}{"9.128..."}\right)$ or $P([Z >] -0.49..)$	M1	3.4
	$= 0.688976...$ awrt 0.689	A1	1.1b
		(4)	
		(11 marks)	
	Notes		
	If you see any attempt using an n-sided die with n not equal to 6 please send to review.		
(a)	M1 for sight or use of the correct distribution. <u>Must</u> have B, or Bin or Bpd or Bcd and the correct value for n and p , just $n = 10, p = \frac{1}{6}$ is M0		
	Implied by one answer correct to 2dp or by sight of $\binom{10}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^7$ or one of :		
	$[P(X = 0) =]$ 0.16 (1...), $[P(X = 1) =]$ 0.32 (3...), $[P(X = 2) =]$ 0.29 (0...), $[P(X \leq 3) =]$ 0.93 (0...)		
(i)	1 st A1 for awrt 0.155		
(ii)	2 nd A1 for awrt 0.775		
(b)	1 st M1 for selecting a correct model. Sight or use of correct binomial, ft their (a)(i) May be implied by sight of $[P(D \leq 11) =]$ 0.78... or 0.79 or $[P(D \leq 12) =]$ 0.87...		
	2 nd M1 for correct interpretation of "at least 12" and writing or using $1 - P(D \leq 11)$		
	We are <u>not</u> attempting to ft their incorrect 0.155 on our calculators here.		
	A1 for awrt 0.212 [Answer only 3/3]		
(c)	B1 for 100 but must be seen in part (c) i.e. between (b) and (d)		
(d)	1 st M1 for attempting normal with mean = 100 or ft their answer to (c) May be implied by the correct mean and a correctly labelled s.d. (σ) or var (σ^2)		
	1 st A1 for correctly labelled standard deviation allow $\sqrt{\frac{250}{3}} = \sqrt{83.3...} = 9.1(28....)$ or correctly labelled variance. Implied by $N(\mu, \frac{250}{3})$ or correct answer.		
	2 nd M1 for attempt at continuity correction i.e. sight of 95 ± 0.5		
	2 nd A1 for awrt 0.689 [Answer only 4/4]		
NB	If they don't state the model for 1st M1 but just give probabilities with probability statements (Y is any letter):		
$\sigma = \frac{250}{3}$	1 st M1 implied by: $P(Y > 94.5) = 0.52(63...)$, $P(Y > 95) = 0.52(39...)$, $P(Y > 95.5) = 0.52(15..)$		
No cc	1 st M1 1 st A1 implied by: $P(T > 95) = 0.70(805...)$		
	1 st M1 1 st A1 2 nd M1 implied by: $P(T > 94.5) = 0.72(657...)$		
	Exact binomial gives 0.68567... and will likely score 0/4		

Qu 2	Scheme	Marks	AO
(a)	e.g. The <u>height</u> (h) <u>decreases</u> by about <u>1.28 m</u> for <u>each second</u> of the flight	B1	3.4
(b)	$H_0 : \rho = 0$ $H_1 : \rho < 0$ [5% 1-tail cv =] ($\pm$) 0.5494 [$r = -0.510$ not sig] there is <u>insufficient</u> (o.e.) evidence of a negative <u>correlation</u> between <u>height</u> (or h) and <u>time</u> (or t)	B1 M1 A1	(1) 2.5 1.1a 2.2b
(c)	No – since points seem to follow a curve/quadratic (rather than a line) or since points are “non-linear” but regression line/ model is linear or e.g. between ($t = 5$ and 7) height drops by much more than 2.56 m or e.g. gradient is positive up to $t = 3.5$ (line gradient < 0) or e.g. gradient is positive initially (line gradient < 0) or e.g. gradient is positive and then negative	B1	2.4
(d)	[$h = 38.1 - 0.78 (t - k)^2$ with] a suitable k i.e. in the range 3~4.5	B1	(1) 3.3
(6 marks)			
Notes			
(a)	B1 for a suitable interpretation in context [value can be 1.3 or 1.28 or “just over 1”] per sec Must have underlined words (o.e.) and units “m” or metres and “s” or seconds NB “descends” implies “height decreases” Condone e.g. “decreases by – 1.28 m”		
(b)	B1 for both hypotheses correct in terms of ρ [accept a p or p but not r or r] Must be attached to H_0 and H_1 M1 for a critical value corresponding to their H_1 : 1-tail: awrt ± 0.549 or 2-tail (B0 scored for H_1) : awrt ± 0.632 (tables 0.6319) If hypotheses are in words and can deduce whether one or two-tail then use their words. If no hypotheses or their H_1 is not clearly one or two-tail assume one-tail A1 a correct conclusion in context mentioning <u>correlation</u> and <u>height</u> and <u>time</u> A comparison or statement such as “not sig” is not needed but if seen must be correct. Do NOT award this A mark if contradictory comments or working seen e.g. “reject H_0 ” or comparison of 0.510 with significance level of 0.05 or e.g. $-0.549 > -0.510$		
NB	Can award B0M1A1		
SC	B0 (for 2-tail) M0 (for cv = ± 0.549) scored: Allow 1 mark (score as B0M0A1) for conclusion such as: “ <u>insufficient</u> evidence of (negative) <u>correlation</u> between <u>height</u> and <u>time</u> of flight”		
(c)	B1 for saying no and giving a suitable supporting reason Don’t allow “correlation” on its own instead of “gradient” B0 for simply saying “points don’t lie close to a straight line” Need mention of curve or some other feature of scatter plot that <u>differs</u> from regression line. B0 for just “non-linear” without mention of the model being linear B0 for simply comparing 1 or 2 points – need a comment about general pattern		
(d)	B1 for a value of k in the range [3, 4.5] Do not need $k = \dots$ Accept a value embedded in Jane’s model. ISW any errors in multiplying out bracket.		

Qu 3	Scheme	Marks	AO
(a)	Rain[fall] (allow [Mean] Windspeed)	B1 (1)	1.2
(b)	$[\bar{x} =] 15.2239... = \text{awrt } \underline{15.2}$ $\sigma_x = \sqrt{\frac{44\,695.4}{184} - "15.22.."^2} \quad \text{or} \quad \sqrt{11.1(422...)}$ $= 3.33800... \quad \text{awrt } \underline{3.34}$	B1 M1 A1 (3)	1.1b 1.1b 1.1b
(c)	<u>Mean is higher</u> than average OR a <u>summer/spring</u> month If they say winter/autumn they must explain that these are hotter months for Perth. [Perth is southern hemisphere or Australia so latest available] month is Oct	M1 A1cso (2)	2.4 2.2b
(6 marks)			
Notes			
Answers may appear next to the question.			
(a)	B1 for Rain[fall] or precipitation (e.g. allow Daily Total [or Mean or max] Rainfall etc) (or allow Mean Windspeed or just “windspeed” BUT not max windspeed or “gust”) If they give more than one answer we take the last one . (NB Actual windspeed mean is 8.2, sd 2.38. No other quantitative variables available)		
(b)	B1 for awrt 15.2 (Do not accept fractions or mixed numbers) M1 for a correct expression including square root (ft their mean) May be implied by an answer of 3.3 or better. A1 for awrt 3.34 [Allow $s = 3.3471...$ i.e. awrt 3.35 if correct formula/expression is seen]		
(c)	If answer in (b)(i) > 19.4 and an attempt is made in (c) please send to review. M1 for a reason mentioning that mean or temperature is higher (o.e.) e.g. it is a <u>warmer/hotter</u> month is OK <u>or</u> sight of $19.4 > (\text{their}) 15.2$ Only ft their 15.2 if it is less than 19.4 OR suggesting a summer/spring month. Ignore incorrect statements that are irrelevant or don’t contradict For incorrect statements that contradict score M0 A1cso dep on M1 scored for inferring October Must choose <u>just</u> October not a range like August~October (NB actual mean for Sep is 15.6 and sd 3.19 and this scores A0) Can accept for example “high mean so December” for M1A0		
SC	M1A1 for “October since Perth is in the southern hemisphere/Australia” M1A0 for “Sep <u>or</u> Nov <u>or</u> Dec <u>or</u> Jan <u>or</u> Feb and “Perth is in the southern hemisphere/Australia” M0A0 just “Perth is in the southern hemisphere/Australia” <u>without</u> a month M0A0		

Qu 4	Scheme	Marks	AO
(a)	$H_0 : p = 0.1$ $H_1 : p \neq 0.1$ [Allow 10% for 0.1]	B1	2.5
	$[X \sim B(40, 0.1)] \Rightarrow P(X = 0) = 0.0148$ [Allow any letter for X]	M1	3.3
	$P(X \geq 9) = 1 - P(X \leq 8) = 1 - 0.9845 = 0.0155$	A1	1.1b
	Critical region is $\underline{\{X = 0\} \cup \{X \geq 9\}}$ (o.e.)	A1	1.1b
		(4)	
(b)	$[“0.0148” + “0.0155”] = 0.0303$	B1ft	1.1b
		(1)	
(c)	[Provided 7 is not in their CR] insufficient evidence to support Freya’s belief	B1	2.2b
		(1)	
(6 marks)			
Notes			
Mark (a) and (b) together for sight of the probabilities.			
(a)	B1 for both hypotheses in terms of p or π. Must be attached to H_0 and H_1 M1 for <u>use</u> of the correct model. Implied by sight of at least one probability truncated or rounded to at least 2sf from: 0.0155, 0.0148, 0.0805, 0.9845, 0.9581, 0.9949 Implied by sight of fully correct CR (with no probs) so e.g. $X = 0$, $X > 8$ scores M1A0A0 1st A1 for at least one correct probability (to at least 3sf) with its probability statement i.e. for $P(X = 0) = \text{awrt } 0.0148$ <u>or</u> $P(X \geq 9) = \text{awrt } 0.0155$ 2nd A1(dep on M1 and 1st A1 but not on B1) for both correct probs (to at least 3 sf) <u>and</u> the correct critical region Do not need set notation. Allow $X < 1$ and $X > 8$ <u>or</u> words e.g. “0 or greater than 8” etc Allow “,” <u>or</u> “and” <u>or</u> “or” <u>or</u> “$\cap$” between $X \leq 0$ and $X \geq 9$ $P(X = 0)$ and $P(X \geq 9)$ is 2nd A0		
(b)	B1ft for awrt 3.03% <u>or</u> correct sum of their two probabilities (provided each is less than 0.5) Their probabilities must be to at least 2sf and relate to their CR To score in (c) they must have a CR of the form $(X = 0 \text{ or } X < a)$ and $X > b$, where b is ≥ 7 May be implied by $P(X < a)$ and $P(X > b)$ i.e. 2nd A0 in (a) but correct form. Need $b > a$		
(c)	B1 for a suitable comment in context that suggests <u>no support</u> for Freya’s <u>belief/claim</u> or e.g. <u>insufficient</u> evidence of change in <u>proportion/percentage</u> of <u>left-handed</u> adults or e.g. <u>proportion/percentage</u> of <u>left-handed</u> adults is <u>not different</u> from 10% (or ...is 10%) or e.g. <u>10%</u> of adults in the country are <u>left-handed</u> Do not allow contradictory comments e.g. “in CR so no support for Freya’s belief” is B0		
NB	A correct contextual answer in (c) using an acceptance region please send to review.		

Qu 5	Scheme	Marks	AO
(a)	$[P(H > 1.6) =] \ 0.091211... = \text{awrt } \underline{0.0912}$	B1 (1)	1.1b
(b)	Need H and T to be independent or events $\{H > 1.6\}$ and $\{T < 300\}$ are independent	B1 (1)	2.4
(c)	$[P(T < 300) =] \ 0.124(2816...)$ Prob both is: $"0.0912..." \times "0.124..."$ $= 0.011335... = \text{awrt } \underline{0.0113}$	M1 M1 A1 (3)	3.4 1.1b 1.1b
(d)	$\frac{16.3 - \mu}{\sigma} = -0.5244(0051...) , \frac{29 - \mu}{\sigma} = 1.2816$ (calc: 1.28155156...) e.g. $29 - 16.3 = \sigma("1.2816" - "-0.5244")$ $\sigma = 7.032115... = \text{awrt } \underline{7.03}$ $\mu = 19.9876... = \underline{19.95} \leq \underline{\mu} \leq \underline{20.0}$	M1M1 M1 A1 A1 (5)	3.1a 1.1b 1.1b 3.2a
(10 marks)			
Notes			
(a)	B1 for awrt 0.0912 (from calculator)		
(b)	B1 for a suitable reason mentioning or implying H and T are independent Allow: e.g. “they”/ “each event”/ “ $P(H)$ and $P(T)$ ”/ “the variables” and “independent” B0 for “the results” / “the values” are independent. Ignore other comments that are not incorrect or contradictory.		
(c)	1 st M1 for using model for T to attempt to find $P(T < 300)$ e.g. sight of 0.124 or better <u>or</u> sight of $\pm \left(\frac{300 - 330}{26} \right)$ or $\pm \left(\frac{5 - 5.5}{0.433...} \right)$ or $Z = \pm 1.15(3...)$ 2 nd M1 for multiplying their two probabilities together ft part (a) and their $P(T < 300)$ provided both values are probabilities. NB M0M1 is possible here A1 for awrt 0.0113 [Correct answer with no incorrect working 3/3]		
(d)	1 st M1 for standardising 16.3 and setting equal to z value where $0.5 < z < 0.6$ 2 nd M1 for standardising 29 and setting equal to z value where $1 < z < 1.5$ 3 rd M1 dep on 1 st or 2 nd M1 for solving their two linear eq'ns – reach an eq'n in one variable May be implied by sight of $\sigma = 7$ (or better) or $\mu = 20$ (or better) For 1st A mark we must also see one of – 0.5244 or 1.2816 (or better) <u>used</u> in their equ'ns OR both z values correct to 3dp i.e. – 0.524 and 1.282 1 st A1 for $\sigma = \text{awrt } 7.03$ (but see 3 rd case below) 2 nd A1 for $\mu = \text{in } [19.95, 20.0]$ (i.e shouldn't see something rounding down to 20.0) Allow 20 from equations with suitable z values (see examples below)		
NB	Use of – 0.524 and 1.28 [would give 7.0399... and 19.988...] and scores M3A0A1 Use of – 0.524 and 1.2816 [would give 7.033... and 19.99...] and scores M3A1A1 Use of – 0.5244 and 1.28 [would give 7.038... and 19.99 ...] and scores M3A1A1 Both z values correct to 3dp i.e. – 0.524 and 1.282 [should give 7.032 and 19.984] scores A1A1		

Qu 6	Scheme	Marks	AO
(a)	A, C <u>or</u> A, D <u>or</u> B, D [Allow things like $A \cap D$]	B1 (1)	1.2
(b)	$P(C) = 0.6$ <u>and</u> $P(B) = p + 0.32$ <u>and</u> $P(B \cap C) = 0.27$ <u>or</u> $(0.08 + 0.25 + 0.27) \times (0.27 + 0.05 + p) = 0.27$ <u>or</u> $0.27 + 0.05 + p = \frac{0.27}{0.6} = 0.45$ [$p + 0.32 = 0.45$ so] $p = \underline{0.13}$	M1 A1 (2)	1.1b 2.2a
(c)	$[P(A B')] = \frac{q}{q+r+0.25+0.08}$ <u>or</u> $\frac{q}{1-(0.05+"0.13"+0.27)}$ <u>or</u> $\frac{q}{0.55}$ $q + r = 1 - 0.65 - "0.13" [= 0.22]$ Since $r \geq 0$ the greatest value of q is "0.22" so $P(A B') \leq \underline{0.4}$ <u>or</u> $\underline{\frac{2}{5}}$	M1 M1 A1 (3)	2.1 1.1b 2.2a
(d)	$[P(B A')] = \frac{0.27+"0.13"}{0.6+"0.13"+r} = 0.5$ <u>or</u> $\frac{0.27+"0.13"}{1-(q+0.05)} = 0.5$ $r = \underline{0.07}$, $q = \underline{0.15}$	M1 A1 A1ft (3)	1.1b 1.1b 1.1b
(e)	$[P((A \cup B)' \cap C)] = [0.25 + 0.08] = \underline{0.33}$	B1 (1)	1.1b
(f)	e.g. $B \cap [A \cup C]'$ <u>or</u> $B \cap A' \cap C'$ <u>or</u> $(B \cap A') \cap (B \cap C')$ o.e.	B1 (1)	1.1b
(11 marks)			
Notes			
(a)	B1 for a correct pair. If more than one pair is given then all must be correct. $P(A)$ and $P(C)$ etc is B0 $P(A \cap C) = 0$ is B0 but condone things like $A \cap C = \emptyset$ In parts (b) – (d) we will condone poor notation and mark equations/expressions		
(b)	M1 for all relevant labelled probabilities listed <u>or</u> a correct equation/expression for p A1 for $p = 0.13$ In parts (c) and (d) they can use letter p or we ft their value for p provided a probability		
(c)	1 st M1 for a correct method for $P(A B')$ in q (and r) ft their p . May be done in stages e.g. find correct expression for $P(B')$, simplify incorrectly then use q over this 2 nd M1 for a correct equation for $q + r$ (o.e.)(ft their p) Can accept $r = 0$ and $q = 0.22$ NB sight of $\frac{0.22}{0.55}$ will score M1M1 A1 for 0.4 i.e. deducing the maximum value of $P(A B')$. Allow ≤ 0.4 or $P(A B') = 0.4$ Can award 3/3 for $P(A B') = 0.4$ but not 0.4 alone as it can come from e.g. $P(C')$		
(d)	M1 for a correct equation for r (or q) only can have p or ft their value for p . May be in stages e.g. find $P(A') = 0.27 + 0.25 + 0.08 + p + r$ but make a slip in getting 0.6 then use this. 1 st A1 for $r = 0.07$ <u>or</u> $q = 0.15$ 2 nd A1ft for $r = 0.07$ <u>and</u> $q = 0.15$ <u>or</u> values giving $q + r = 0.22$ provided both q and r are probabilities. Obviously, 2 nd A1ft is dependent on the M1		
(e)	B1 for 0.33		
(f)	B1 for any correct expression. Do not condone $P(\dots$		



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Question Paper Log Number P75696A

Publications Code 9MA0_32_2406_MS

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 5. Where a candidate has made multiple responses and indicates which response they wish to submit, examiners should mark this response.
If there are several attempts at a question which have not been crossed out, examiners should mark the final answer which is the answer that is the most complete.
 6. Ignore wrong working or incorrect statements following a correct answer.
 7. Mark schemes will firstly show the solution judged to be the most common response expected from candidates. Where appropriate, alternatives answers are provided in the notes. If examiners are not sure if an answer is acceptable, they will check the mark scheme to see if an alternative answer is given for the method used.

General Principles for Mechanics Marking

(But note that specific mark schemes may sometimes override these general principles)

- Rules for M marks: correct no. of terms; dimensionally correct; all terms that need resolving (i.e. multiplied by cos or sin) are resolved.
- Omission or extra g in a resolution is an accuracy error not method error.
- Omission of mass from a resolution is a method error.
- Omission of a length from a moments equation is a method error.
- Omission of units or incorrect units is not (usually) counted as an accuracy error.
- DM indicates a dependent method mark i.e. one that can only be awarded if a previous specified method mark has been awarded.
- Any numerical answer which comes from use of $g = 9.8$ should be given to 2 or 3 SF.
- Use of $g = 9.81$ should be penalised once per (complete) question.

N.B. Over-accuracy or under-accuracy of correct answers should only be penalised *once* per complete question. However, premature approximation should be penalised every time it occurs.

- Marks must be entered in the same order as they appear on the mark scheme.
- In all cases, if the candidate clearly labels their working under a particular part of a question i.e. (a) or (b) or (c),.....then that working can only score marks for that part of the question.
- Accept column vectors in all cases.
- Misreads – if a misread does not alter the character of a question or materially simplify it, deduct two from any A or B marks gained, bearing in mind that after a misread, the subsequent A marks affected are treated as A ft
- Mechanics Abbreviations

M(A) Taking moments about A.

N2L Newton's Second Law (Equation of Motion)

NEL Newton's Experimental Law (Newton's Law of Impact)

HL Hooke's Law

SHM Simple harmonic motion

PCLM Principle of conservation of linear momentum

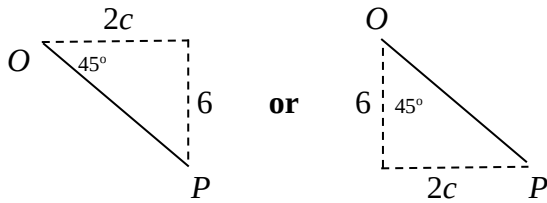
RHS, LHS Right hand side, left hand side.

Question	Scheme		Marks	AOs
1(a)	0.5g, $\frac{1}{2}g$ or 4.9 (N) seen		B1	3.4
			(1)	
1(b)	$\frac{2}{7} \times 4.9$ oe seen		M1	3.1a
	1.4, 1.40 or $\frac{1}{7}g$		A1	1.1b
			(2)	
(3 marks)				
Notes: Ignore units in this question. N.B. Use of $g = 9.81$ should only be penalised once for the whole question as should two fractional answers (a) 49/10 (b) 7/5. Penalise the use of $g = 9.81$ the first time you see it. N.B. If $g = 9.81$ is used in (a), B0. If it is then used again in part (b), and they give the answer as 1.4 or 1.40, they can score M1A1 in part (b). If $g = 9.81$ is only used in (b), they can score max M1A0 for (b).				
1a	B1	cao. (must be positive) B0 for a fraction ($\frac{49}{10}$) B0 if they have 0.5g and then clearly use $g = 9.81$ i.e. NOT isw This answer must appear in (a) to earn this mark. If no labelling, give BOD and award as many marks as possible.		
1b	M1	$\frac{2}{7} \times$ their (a) (must be a numerical value) If no answer for (a) or if (a) is incorrect and they don't use it, allow a correct restart i.e. $\frac{2}{7} \times 4.9$ or $\frac{2}{7} \times 0.5g$ or $\frac{2}{7} \times 0.5$ (missing g is not an M error)		
	A1	A0 for a fraction N.B. $X =$ is not needed BUT A0 for F (or P or horizontal force) = a correct answer if they don't go on to state $X = \dots$ If they obtain the correct answer for F and they have said $F = X$ they can score A1.		
		N.B. 1.4, 1.40 or $\frac{1}{7}g$ with no working , scores M1A1		

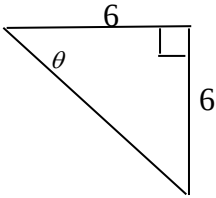
Question	Scheme	Marks	AOs
2(a)	$\frac{10}{4}$	M1	3.4
	$2.5, \frac{5}{2}, \frac{10}{4} \text{ m s}^{-2}$ units needed.	A1	1.1b
		(2)	
2(b)	Find the area, with correct structure, from $t = 0$ to 18	M1	3.1b
	$\frac{1}{2} \times 4 \times 10 + (14 \times 10)$ triangle + rectangle or $\frac{1}{2} \times 10 \times (14 + 18)$ trapezium or $(18 \times 10) - \frac{1}{2} \times 4 \times 10$ rectangle – triangle N.B. $\frac{1}{2} \times 4 \times 10$ may be replaced by $\frac{1}{2} \times 2.5 \times 4^2$ using $s = ut + \frac{1}{2}at^2$ or by $\frac{10^2 - 0^2}{2 \times 2.5}$ using $v^2 = u^2 + 2as$	A1	1.1b
	160 (m)	A1	1.1b
		(3)	
2(c)	Using area, from $t = 18$ to $t = 24$, = (200 – their (b)) with correct structure OR $s = (200 - \text{their (b)})$, using <i>suvat</i> to find s N.B. If their (b) is incorrect and they don't use it, allow a correct restart.	M1	3.1b
	$6U + \frac{1}{2} \times 6 \times (10 - U) = 200 - \text{their (b)}$ rectangle + triangle or $\frac{1}{2} \times 6 \times (10 + U) = 200 - \text{their (b)}$ trapezium ($s = \left(\frac{u+v}{2}\right)t$) or $(6 \times 10) - \frac{1}{2} \times 6 \times (10 - U) = 200 - \text{their (b)}$ rectangle – triangle or $(10 \times 6) + \frac{1}{2} \left(-\frac{(10 - U)}{6}\right) \times 6^2 = 200 - \text{their (b)}$ $s = ut + \frac{1}{2}at^2$ or $(U \times 6) - \frac{1}{2} \left(-\frac{(10 - U)}{6}\right) \times 6^2 = 200 - \text{their (b)}$ $s = vt - \frac{1}{2}at^2$ N.B. Two stage <i>suvat</i> method: $(10 \times 6) + \frac{1}{2}a \times 6^2 = 200 - \text{their (b)} \Rightarrow$ AND $U = 10 + 6 \times \text{their } a$	A1ft	1.1b

		$\frac{10}{3} = 3\frac{1}{3}$ oe	A1	1.1b
			(3)	
(8 marks)				
Notes:				
2a	M1	Any complete <i>suvat</i> method to find <i>a</i> e.g. use $s = 20$ and $20 = \frac{1}{2}a \times 4^2$ N.B. Ignore units at this stage		
	A1	Any equivalent number with correct units. Accept m/s^2 , m/s/s , m per s per s .		
2b	M1	Complete method, they may use <i>suvat</i> on one or more sections, to find the TOTAL area. M0 if a single <i>suvat</i> equation is used for the whole motion M0 if $\frac{1}{2}$ not seen used in an area method		
	A1	Correct unsimplified expression.		
	A1	cao. Ignore units. N.B. Correct answer, with no working , can score all 3 marks.		
2c	M1	Complete method, using area or <i>suvat</i> , to give an equation in <i>U</i> only, with correct structure M0 if $\frac{1}{2}$ not seen used in an area method M0 if 10 is used instead of $(10 - U)$ or $(10 - U)$ is used instead of $(10 + U)$ in any equation		
	A1ft	Correct unsimplified equation in <i>U</i> only (allow <i>V</i> or <i>v</i> instead of <i>U</i>), ft on their 160 .		
	A1	Accept 3.3 or better. Ignore units. Allow use of <i>V</i> throughout instead of <i>U</i> , <u>including in the answer</u> . N.B. Correct answer, with no working , can score all 3 marks.		

Question		Scheme	Marks	AOs
3(a)		$(R =) mg \cos \alpha$	M1	3.4
		$= \frac{12}{13} mg$	A1	1.1b
			(2)	
3(b)		Equation of motion down the plane	M1	2.1
		$mg \sin \alpha - F = ma$ or $mg \sin \alpha - F = -ma$	A1	1.1b
		$(F =) \mu \times \text{their } R$	M1	3.4
		$\frac{1}{13} g(5 - 12\mu) *$	A1*	2.2a
			(4)	
3(c)		P wouldn't move	B1	2.4
			(1)	
(7 marks)				
Notes:				
3a	M1	Correct no. of terms, condone sin/cos confusion and sign errors, dimensionally correct. Allow use of a different symbol for the angle.		
	A1	Accept $0.92mg$ or better. Must be positive.		
3b	M1	Correct no. of terms, condone sin/cos confusion and sign errors (M0 if they use g for a) N.B. Must be using mg or m not W for the weight.		
	A1	Any correct equation e.g. $mg \sin \alpha = ma + F$ N.B. F does not need to be substituted.		
	M1	$\mu \times \text{their } R$ seen (possibly on a diagram), any trig does not need to be replaced. M0 if they use $\mu = \frac{5}{12}$ M0 for just μR with R not replaced.		
	A1*	Given answer correctly obtained, with at least one further line of working with both trig ratios substituted as fractions. Allow $\frac{g}{13}(5 - 12\mu)$ or $\frac{g(5 - 12\mu)}{13}$		
3c	B1	e.g. P (or the particle or the object) would stay at rest, P would not slide down, would not roll down the plane, static equilibrium, equilibrium at rest Allow 'it' for P or just 'stays at rest oe' B0 for wouldn't accelerate, would be in equilibrium (only), would stop N.B. Ignore reasons but not contradictions.		

Question	Scheme	Marks	AOs
4(a)	ALTERNATIVES when $t = 4$ is substituted at the beginning.		
	$2c\mathbf{i} - 6\mathbf{j}$ or as a column vector, seen or implied.	B1	1.1b
	ALT 1  AND either $\tan 45^\circ = \frac{2c}{6} \Rightarrow 2c = 6$ or states isosceles triangle so $2c = 6$ N.B. In both of the above, we must see the justification for the equation. ALT 2 $\tan 135^\circ = \frac{2c}{-6} \Rightarrow 2c = 6$ N.B. M0 if they are using the wrong bearing.	M1	3.1a
	$c = 3 *$	A1*	2.2a
	SC 1 M1A0: no right-angled triangle $2c\mathbf{i} - 6\mathbf{j} = k(\mathbf{i} - \mathbf{j}) \Rightarrow 2c = 6$ or $\mathbf{i}\text{-cpt} = -\mathbf{j}\text{-cpt} \Rightarrow 2c = 6$ N.B. In both of the above, we must see the justification for the equation. SC 2 M1A0: no right-angled triangle $\tan 45^\circ = \frac{2c}{6} \text{ or } \frac{6}{2c} \Rightarrow 2c = 6$ N.B. In the above, we must see the justification for the equation.		

	ALTERNATIVES when $t = 4$ is substituted at the end:		
	$ct^{\frac{1}{2}} = 2c$ and $(-)\frac{3t^2}{8} = (-)6$ when $t = 4$, seen or implied	B1	
	ALT 3 AND either $\tan 45^\circ = \frac{\frac{3t^2}{8}}{ct^{\frac{1}{2}}} \Rightarrow 2c = 6$ when $t = 4$ or states isosceles triangle, so $ct^{\frac{1}{2}} = \frac{3t^2}{8} \Rightarrow 2c = 6$ when $t = 4$ N.B. In both of the above, we must see the justification for the equation. N.B. M0 if they are using the wrong bearing.	M1	
	$c = 3$	A1*	
	SC 3 M1A0: no right-angled triangle $(ct^{\frac{1}{2}}\mathbf{i} - \frac{3t^2}{8}\mathbf{j}) = k(\mathbf{i} - \mathbf{j}) \Rightarrow 2c = 6 \text{ when } t = 4$ or $\mathbf{i}\text{-cpt} = -\mathbf{j}\text{-cpt} \Rightarrow ct^{\frac{1}{2}} = \frac{3t^2}{8} \Rightarrow 2c = 6 \text{ when } t = 4$ N.B. In both of the above, we must see the justification for the equation. SC 4 M1A0: no right-angled triangle $\tan 45^\circ = \frac{\frac{3t^2}{8}}{ct^{\frac{1}{2}}} \Rightarrow 2c = 6 \text{ when } t = 4$.		

	N.B. Allow a verification: i.e. use $c = 3$ and $t = 4$ to show that P is on a bearing of 135° from O. $6\mathbf{i} - 6\mathbf{j}$ then a diagram:  AND $\tan \theta = \frac{6}{6}$ or isosceles triangle $\Rightarrow \theta = 45^\circ$ N.B. In the above, we must see the justification for the equation.	B1	
	bearing = $45^\circ + 90^\circ = 135^\circ$	A1*	
		(3)	
4(b)	Differentiate $\mathbf{r}$ wrt t to obtain $\mathbf{v}$	M1	2.1
	$\mathbf{v} = 3 \times \frac{1}{2} t^{-\frac{1}{2}} \mathbf{i} - \frac{3}{8} \times 2t \mathbf{j} = \frac{3}{2} t^{-\frac{1}{2}} \mathbf{i} - \frac{3}{4} t \mathbf{j}$ oe	A1	1.1b
	Put $t = 4$ into both components and use Pythagoras: $\sqrt{\left(\frac{3}{4}\right)^2 + (-3)^2}$	M1	3.1a
	$\sqrt{\frac{153}{16}}$ or $\frac{\sqrt{153}}{4}$ or $\frac{3\sqrt{17}}{4}$ or $3\sqrt{\frac{17}{16}} = 3.0923..$ (m s ⁻¹)	A1	1.1b
		(4)	
4(c)	Differentiate their $\mathbf{v}$ wrt t to obtain $\mathbf{a}$	M1	3.4
	$\mathbf{a} = -\frac{3}{4} t^{-\frac{3}{2}} \mathbf{i} - \frac{3}{4} \mathbf{j}$	A1	1.1b
	$\frac{-\frac{3}{4} T^{-\frac{3}{2}}}{-\frac{3}{4}} = \frac{-1}{-27}$ oe	M1	2.1
	($T =$) 9	A1	1.1b
		(4)	

(11 marks)		
Notes: Accept column vectors throughout		
4a	B1	$2c\mathbf{i} - 6\mathbf{j}$ seen or implied. B0 for $r = 2c - 6$ if no evidence of components.
	M1	ALT 1: Use the bearing to obtain a CORRECT diagram showing a right-angled triangle with at least one 45° angle marked or clearly explained (i.e. 135° marked on the diagram and either $135^\circ - 90^\circ = 45^\circ$ or $180^\circ - 135^\circ = 45^\circ$), and $2c$ and ± 6 marked AND use of isosceles triangle or tan or (sin/cos and Pythag) to obtain $2c = 6$ ALT 2: No diagram required Use $\tan 135^\circ = \frac{2c}{-6} \Rightarrow 2c = 6$
	A1*	Given answer correctly obtained
		ALTERNATIVE when $t = 4$ is substituted at the end:
	B1	$ct^{\frac{1}{2}} = 2c$ and $(-)\frac{3t^2}{8} = (-)6$ when $t = 4$, seen or implied
	M1	ALT 3: Use the bearing to obtain a CORRECT diagram showing a right-angled triangle with at least one 45° angle marked or clearly explained (i.e. 135° marked on the diagram and either $135^\circ - 90^\circ = 45^\circ$ or $180^\circ - 135^\circ = 45^\circ$), and $ct^{\frac{1}{2}}$ and $\pm \frac{3t^2}{8}$ marked AND use of isosceles triangle or tan or (sin/cos and Pythag) to obtain $2c = 6$ when $t = 4$
	A1*	Given answer correctly obtained
4b	M1	Both powers of t decreasing by 1 (M0 if $\mathbf{i}$ or $\mathbf{j}$ is missing but allow recovery or working with components only) N.B. This mark is available if c has not been substituted for.
	A1	Correct unsimplified derivative or two correct components
	M1	Put $t = 4$ in their $\mathbf{v}$ (must be using an attempted derivative of $\mathbf{r}$) and then use Pythagoras with the root, allow a missing $-$ sign N.B. If they state $t = 4$, allow a slip when they substitute in, for this M mark. This mark is available if c has not been substituted for.
	A1	Accept 3.1 or better
4c	M1	Both powers of t decreasing by 1 N.B. This mark is available if c has not been substituted for (M0 if $\mathbf{i}$ or $\mathbf{j}$ is missing but allow recovery or working with components only).
	A1	Correct unsimplified derivative
	M1	Use of an appropriate ratio (must be using an attempted derivative of their v), condone sign error and the reciprocal , to obtain an equation in t or T only . N.B. If they state that $-\frac{3}{4}T^{-\frac{3}{2}}\mathbf{i} - \frac{3}{4}\mathbf{j} = k(-\mathbf{i} - 27\mathbf{j})$ and then equate coefficients to give two simultaneous equations in k and T , these need to be used to produce an equation in T only , before the M mark is earned.
	A1	cao (allow t instead of T)

Question	Scheme	Marks	AOs
5(a)	Using horizontal motion, $s = ut$, with 35 resolved	M1	3.3
	$x = 35 \cos \alpha \times t$	A1	1.1b
	Using vertical motion, $s = ut + \frac{1}{2}at^2$, with 35 resolved	M1	3.4
	$y = 35 \sin \alpha \times t - \frac{1}{2}gt^2$	A1	1.1b
	Eliminate t : $y = 35 \sin \alpha \times \frac{x}{35 \cos \alpha} - \frac{1}{2}g \left(\frac{x}{35 \cos \alpha} \right)^2$	DM1	3.1b
	$y = \frac{3}{4}x - \frac{1}{160}x^2$ *	A1*	1.1b
	N.B. No marks available if they just quote the equation of the path.		
5(b)	ALTERNATIVE: they do (b) and/or (c) first <u>using a suvat method</u> Assume $y = ax^2 + bx + c$ Use any three of (0,0), (120,0) from part (b), (60,22.5) from part (c) or $\frac{dy}{dx} = \frac{3}{4}$ at $x = 0$ to find a, b, and c. M1A1, M1A1, DM1A1* for finding each of a, b and c and stating final answer in correct form. N.B. If they realise that $c = 0$, and just use $y = ax^2 + bx$, that could score M1A1. Enter marks on ePEN in the order in which a, b and c are found: e.g. $x = 0, y = 0 \Rightarrow c = 0$ M1A1 $x = 0, \frac{dy}{dx} = 2ax + b = \frac{3}{4} \Rightarrow b = \frac{3}{4}$ M1 A1 $x = 120, y = 0 \Rightarrow 0 = 120^2 a + 120 \times \frac{3}{4} \Rightarrow a = \frac{-1}{160}$ DM1 so, $y = \frac{3}{4}x - \frac{1}{160}x^2$ A1*		
		(6)	
5(b)	ALT 1 $0 = \frac{3}{4}x - \frac{1}{160}x^2$ and solve for x	M1	3.1b
	ALT 2 $\frac{dy}{dx} = \frac{3}{4} - \frac{x}{80} = 0 \Rightarrow x = 60$ and $OA = 2 \times 60$		
	ALT 3		

	A complete <i>suvat</i> method to find OA: e.g. $0 = 35 \sin \alpha \times t - \frac{1}{2}gt^2$ or $0 = 35 \sin \alpha - g \frac{t}{2}$ or $-35 \sin \alpha = 35 \sin \alpha - gt$ to find $t \left(= \frac{70 \sin \alpha}{g} = \frac{30}{7} \right)$ AND $(OA =) 35 \cos \alpha \times t = 35 \cos \alpha \times \frac{70 \sin \alpha}{g}$ N.B. OR use the calculator to input the equation of the path which then gives $y_{\max} = 45/2$ when $x = 60$ with no working, so $OA = 2 \times 60$		
	(OA =) 120 (m)	A1	1.1b
		(2)	
5(c)	ALT 1 $H = \frac{3}{4} \times 60 - \frac{1}{160} \times 60^2$ ALT 2 $y = \frac{-1}{160}(x^2 - 120x) = 22.5 - \frac{1}{160}(x - 60)^2 \quad \text{so max } y = 45/2 \text{ or } 22.5$ ALT 3 $\frac{dy}{dx} = \frac{3}{4} - \frac{2x}{160} = 0 \Rightarrow x = 60 \quad \text{then find } y \text{ when } x = 60$ ALT 4 A complete <i>suvat</i> method: e.g. $H = \frac{(35 \sin \alpha)^2}{2g}$ or $0 = 35 \sin \alpha - gt$ to find the time to top, $t = \frac{35 \sin \alpha}{g}$, or use half their time they found in (b) AND $H = 35 \sin \alpha \times \frac{35 \sin \alpha}{g} - \frac{1}{2}g \left(\frac{35 \sin \alpha}{g} \right)^2 \quad \text{or} \quad \left(\frac{35 \sin \alpha + 0}{2} \right) \times \frac{35 \sin \alpha}{g}$ N.B. OR use the calculator to input the equation of the path which then gives $y_{\max} = 45/2$ (when $x = 60$) with no working.	M1	3.1b
	(H =) 22.5 Accept 23	A1	1.1b

		(2)	
5(d)	H is greater (or K is smaller), as air resistance would slow the particle down oe.	B1	3.5a
		(1)	
5(e)	e.g. the inaccuracy of using 9.8 m s^{-2} for g	B1	3.5b
		(1)	
(12 marks)			
Notes:			
5a	M1	Correct terms but condone sin/cos confusion and sign errors Available if they use s instead of x	
	A1	Correct equation in x and t N.B. they may have the wrong value for $\cos \alpha$	
	M1	Correct terms but condone sin/cos confusion and sign errors Available if they use a different letter for y provided it's not the same as they've used for x . N.B. M0 if they subsequently use a value for y e.g. 0	
	A1	Correct equation in y and t N.B. they may have the wrong value for $\sin \alpha$ They may have t in terms of x , from their first equation.	
	DM 1	Dependent on the two previous M marks for eliminating t to give an equation in y and x only.	
	A1*	Given answer correctly obtained, with at least one further line of working with trig ratios and $g = 9.8$ explicitly seen or 4.9 oe used for 0.5g.. Allow $y = \frac{3x}{4} - \frac{x^2}{160}$ and with y on the RHS	
5b	M1	ALT 1: Use of $y = 0$ in equation of path and solve for x . ALT 2: Use calculus to find the x -coordinate of the max point and double it ALT 3: Any other complete <i>suvat</i> method to find OA , condone sin/cos confusion and sign errors	
	A1	cao	
5c	M1	ALT 1: Use of $x = \left(\frac{1}{2} \times \text{their } OA \right)$ in equation of path ALT 2: Complete the square for the equation of the path and deduce maximum value of y : Need to see $p - q(x - r)^2$ with p as the answer ALT 3: Use $\frac{dy}{dx} = 0$ to find the x -coordinate of the max point and then use the path equation to find the y -coordinate ALT 4: Any complete <i>suvat</i> method to find H , condone sin/cos confusion and sign errors	

	A1	A0 for $\frac{45}{2}$
5d	B1	Possible justifications for H greater (or K smaller): air resistance will provide an extra force acting against the stone or opposing the motion air resistance would take away energy from the stone work would be done against air resistance air resistance will mean the acceleration will be less (or deceleration greater) air resistance would reduce the velocity/speed Allow 'it' for air resistance B0 if no justification given. B0 for air resistance would reduce the initial velocity of the stone B0 for air resistance will limit the vertical height (not a reason) or oppose the vertical force. Ignore extras.
5e	B1	Any single correct answer. Acceptable answer: the inaccuracy of using 9.8 m s^{-2} for g or wind or weather effects or the spin of the stone or the size (or shape or surface area) of the stone or or the stone is still modelled as a particle N.B. Allow if the stone is referred to as e.g. a ball B0 if any incorrect extras Unacceptable answers: The model (it) does not take account of: the mass or weight of the stone air resistance. The ground may not be horizontal. B0 for consequences of air resistance being included e.g. the path won't be a parabola the path won't be symmetrical

Question	Scheme	Marks	AOs
6.			
6(a)	Take moments about A	M1	3.1a
	$S \times 1.5a = Mga \cos \theta = (Mga \times \frac{3}{5})$	A1	1.1b
	$S = \frac{2}{5} Mg^*$	A1*	2.2a
		(3)	
6(b)	N.B. Marks for the equations should be awarded in the order in which they appear on the script.		
	Resolve horizontally:	M1	3.4
	$F = S \sin \theta$	A1	1.1b
	Resolve vertically:	M1	3.3
	$R = Mg - S \cos \theta$	A1	1.1b
	Other possible equations: (any of which is worth max M1A1) (parallel to the rod): $F \cos \theta + R \sin \theta = Mg \sin \theta$ (perp to the rod): $F \sin \theta + Mg \cos \theta = S + R \cos \theta$ M(B): $(S \times 0.5a) + (R \times 2a \cos \theta) = (Mg \times a \cos \theta) + (F \times 2a \sin \theta)$ M(C): $(R \times 1.5a \cos \theta) = (Mg \times 0.5a \cos \theta) + (F \times 1.5a \sin \theta)$ M(G): $(R \times a \cos \theta) = (S \times 0.5a) + (F \times a \sin \theta)$ N.B. If they have more than two equations, mark only those that they use to try to find μ		
	$F = \mu R$ and two of their equations used to solve for μ	DM1	3.1a
	$\mu = \frac{8}{19} = 0.42105\dots$	A1	2.2a

		(6)	
(9 marks)			
Notes:			
6a	M1	Correct no. of terms, dimensionally correct, condone sin/cos confusion and sign errors. Allow use of a different letter for the angle. N.B. They may resolve the weight into two components, parallel and perpendicular to the rod, and then take the moment of each about A, one of which is 0. (see N.B. below) N.B. M0 if one or both a 's aren't there originally.	
	A1	Correct equation, $\cos \theta$ may or may not be replaced by $\frac{3}{5}$ N.B. you may see: $S \times 1.5a = Mg \cos \theta \times a$	
	A1*	Given answer correctly obtained, need to see $\cos \theta = \frac{3}{5}$ used Allow: $S = \frac{2Mg}{5}$ or $\frac{2Mg}{5} = S$ A0 if S is missing	
6b	M1	Correct no. of terms, dimensionally correct, condone sin/cos confusion and sign errors	
	A1	Correct first equation, S does not need to be substituted	
	M1	Correct no. of terms, dimensionally correct, condone sin/cos confusion and sign errors	
	A1	Correct second equation, S does not need to be substituted	
	DM 1	Dependent on previous two M marks for using $F = \mu R$ and two equations to solve for μ	
	A1	Accept 0.42 or better (as g cancels)	

